

From Absolute Scores to Pairwise Preferences: Training LLM Judges for Chinese Text Rewriting Evaluation

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Abstract

Evaluating Chinese text rewriting is inherently subjective: a good rewrite must preserve meaning while improving expression, and annotators often make relative rather than calibrated absolute judgments. This raises a central question: what supervision signal should be used to train LLM-based rewrite judges? We compare absolute-score supervised fine-tuning (SFT) with pairwise preference training for Chinese rewrite quality evaluation. Using 730 human-rated Chinese source–rewrite pairs from three trained annotators ($\text{ICC}(2,1)=0.87$), we construct pairwise preference supervision from holistic human ratings and train RewriteJudge to compare and rank rewrites. Pairwise preference training achieves the highest ranking alignment ($\rho = 0.688$), outperforming absolute-score SFT ($\rho = 0.645$) and substantially exceeding zero-shot and general-purpose LLM judges. Data efficiency analysis shows that pairwise supervision maintains stable performance with only 25% of training data. We further show that source-similarity metrics are negatively aligned with human judgments, confirming that rewrite evaluation requires learning human preferences rather than measuring source overlap. Our results suggest that contrastive preference supervision is a more suitable training paradigm for subjective rewriting evaluation than absolute-score calibration.

1 Introduction

A rewrite should be semantically faithful but expressively different. We focus on *general-purpose Chinese text improvement rewriting*, where the goal is to improve clarity, fluency, and readability while preserving the source meaning. Evaluating such subjective transformations is challenging: annotators often

make relative quality judgments rather than calibrated absolute scores, and standard metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) rely on source overlap, which can be negatively aligned with rewrite quality. This raises a central question: *how should we train LLM-based judges for subjective rewriting evaluation, where absolute score calibration is difficult but relative preference is more natural?*

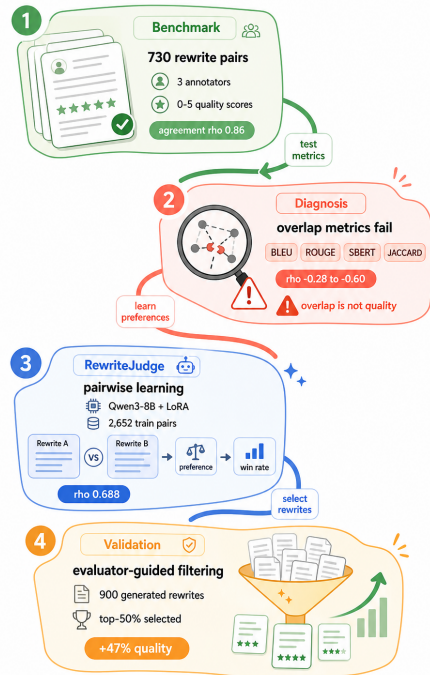


Figure 1: Overview of our approach: we collect human-rated Chinese source–rewrite pairs, construct pairwise preference supervision from holistic human ratings, and train RewriteJudge via contrastive preference learning with win-rate aggregation.

We study two training paradigms for LLM-based rewrite judges: *absolute-score SFT*, which trains the model to predict a 0–5 score directly, and *pairwise preference train-*

ing, which trains the model to compare candidate rewrites and aggregates pairwise preferences into a global ranking. Using 730 human-rated Chinese source-rewrite pairs (Figure 1), we show that pairwise preference training better aligns with human rankings than absolute-score SFT and substantially outperforms zero-shot and general-purpose LLM judges. We further provide diagnostic evidence that source-similarity metrics are negatively aligned with human judgments, motivating the need for learned preference-based evaluation.

Our contributions are:

1. We formulate Chinese rewriting evaluation as a subjective preference-alignment problem, where relative quality judgments are more natural than calibrated absolute scores.
2. We propose RewriteJudge, a pairwise preference training framework that converts human holistic ratings into comparison supervision and aggregates model preferences into win-rate rankings.
3. We show that pairwise preference training achieves stronger alignment with human rankings than absolute-score SFT and substantially outperforms zero-shot and general-purpose LLM judges; diagnostic analyses further explain why source similarity is an unreliable proxy.

2 Related Work

Traditional rewriting evaluation relies on overlap metrics like BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), while embedding metrics such as BERTScore (Zhang et al., 2020), MoverScore (Zhao et al., 2019), BARTScore (Yuan et al., 2021), and SBERT (Reimers and Gurevych, 2019) improve semantic matching through contextualized representations. Methods such as ParaScore (Lan and Chen, 2023) and Alam et al. (2024) consider paraphrase quality and multiple dimensions, and prior surveys document broader obstacles in NLG evaluation (Gehrmann et al., 2023; Gao et al., 2025). These methods mostly treat similarity as a useful proxy for quality. We do not claim to exhaust all automatic evaluation methods; instead, we focus on a diagnostic question: whether source-rewrite similarity, a common

proxy in rewriting evaluation, aligns with human quality judgments.

The LLM-as-a-Judge paradigm offers a flexible alternative to reference metrics. G-Eval (Liu et al., 2024) uses prompting for structured evaluation, Prometheus 2 (Kim et al., 2024) fine-tunes a 7B model for absolute and pairwise scoring, and Chatbot Arena (Chiang et al., 2024) leverages human pairwise comparisons. However, general-purpose judges can exhibit bias and rating inconsistency (Wang et al., 2024a,b; Lee et al., 2025). We examine whether such judges transfer to the specialized, subjective task of Chinese rewrite quality evaluation and whether task-specific training is necessary.

Pairwise comparison is often more stable than absolute scoring because it asks for a relative judgment rather than global calibration. Kim et al. (2024) find higher agreement with human judgments in general NLP tasks, Zheng et al. (2023) use pairwise comparison for LLM assessment, and work on generative selection shows that comparative scoring can improve selection from N candidates (Wang et al., 2024c). Preference learning work further shows that models can learn from pairwise signals without explicit reward scores (Rafailov et al., 2023; Lambert et al., 2024). Prior work has studied LLM judges and pairwise evaluation, but less is known about which supervision signal is better for training task-specific judges in subjective Chinese rewriting evaluation.

3 Preference-Based Rewrite Judge

Given a Chinese source text s and a candidate rewrite r , the evaluator should judge whether r improves expression while preserving the source meaning. A high-quality rewrite should be faithful, fluent, natural, and appropriately transformed. We study two paradigms for training LLM-based judges (Figure 2).

3.1 Human Preference Standardization

We define text rewriting quality along four complementary dimensions, scored from 0 to 5: **Meaning preservation** (preserve the source meaning without factual errors or omissions of key content), **Expression improvement**

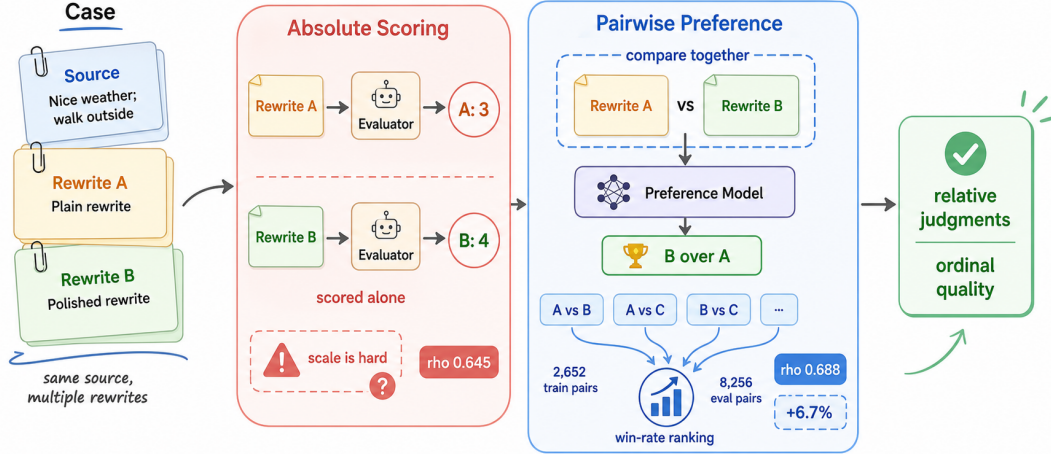


Figure 2: Absolute scoring requires global score calibration across all quality levels; pairwise training learns relative preference and aggregates comparisons into rankings via win rates. Our pairwise setting compares source–rewrite pairs across different sources for global quality ranking.

(improve fluency, clarity, and readability), **Appropriate transformation** (make necessary and effective structural, phrasing, and syntactic adjustments, rather than mechanical substitution or excessive rewriting), and **Naturalness and idiomacity** (ensure the Chinese expression is natural and contextually appropriate). The final score is a holistic judgment integrating all dimensions; annotators provide only a unified rating, using the four dimensions as internal guidelines. We do not reward divergence for its own sake: a rewrite that is lexically or structurally different but semantically distorted or less natural should receive a low score.

We collected 730 Chinese rewrite pairs (730 unique source texts). Source texts contain 49 to 251 Chinese characters (mean 103.2, median 93), while rewrites contain 20 to 247 characters (mean 93.4, median 84.5). Three trained annotators independently scored each pair on the holistic scale from 0 to 5. Agreement is high across multiple statistics: mean pairwise Spearman $\rho = 0.88$, ICC(2,1)=0.87, Krippendorff’s alpha (ordinal)=0.87, mean weighted kappa (quadratic)=0.87, exact agreement=81.1%, and ± 1 agreement=94.1%. We use the average rating as the gold standard. The dataset is split into 600 training pairs and 130 evaluation pairs using stratified sampling by score. The evaluation set closely reflects the full score distribution: 0 (15.5%), 1 (27.9%), 2 (24.8%), 3 (20.2%), 4 (9.3%), 5 (2.3%). Because the

score distribution is imbalanced, we oversample minority score classes and create a balanced training set of 1,008 samples (168 per class). Without balancing, 82% of predictions collapse to Score 1 ($\rho = 0.253$); after balancing, the model predicts across the full score range.

3.2 Absolute Scoring and Pairwise Preference Training

Absolute-score SFT. The standard approach trains a model to predict a holistic quality score:

$$f_{\theta} : \mathcal{S} \times \mathcal{R} \rightarrow [0, 5] \quad (1)$$

The training objective minimizes the discrepancy between predicted and human-assigned scores:

$$\mathcal{L}_{\text{SFT}} = \mathbb{E}_{(s,r,q)} [\ell(f_{\theta}(s,r), q)] \quad (2)$$

where q is the human quality rating and ℓ is a cross-entropy loss over discrete score classes.

Pairwise preference training. Rather than predicting absolute scores, we train the model to compare pairs of rewrites:

$$g_{\theta} : (\mathcal{S} \times \mathcal{R})^2 \rightarrow \{0, 1\} \quad (3)$$

Given two source–rewrite pairs (s_i, r_i) and (s_j, r_j) with human ratings $q_i > q_j$, the model learns:

$$\mathcal{L}_{\text{pair}} = \mathbb{E}_{(r_i, r_j)} [-\log P_{\theta}(r_i \succ r_j \mid q_i > q_j)] \quad (4)$$

The key difference is that absolute scoring requires global calibration across all quality levels, while pairwise training only requires local relative judgments. For subjective tasks where annotators naturally make comparative rather than calibrated judgments, pairwise supervision provides a more natural learning signal.

We fine-tune Qwen3 8B (Yang et al., 2025) and Qwen2.5 7B Instruct (Team, 2025) with LoRA (Hu et al., 2022). For a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA reparameterizes the update as:

$$W' = W_0 + \frac{\alpha}{r}BA \quad (5)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only A and B are trained while W_0 remains frozen. We inject trainable matrices into seven target modules: query, key, value, output projection, and the three MLP layers (gate, up, and down projections), resulting in trainable parameters totaling approximately 0.2% of the full model. For absolute scoring, the model outputs a single holistic score from 0 to 5 using a rubric-grounded prompt. We compare three prompt variants: **Simple** (minimal instructions), **Standard** (brief criteria), and **Detailed** (matching the human annotation rubric; see Appendix C). Rubric-aligned prompts improve absolute scoring ($\rho = 0.645$ vs. $\rho = 0.607$ for Standard), and reasoning prefixes hurt performance ($\rho = 0.360$). For pairwise preference learning, the model receives both source–rewrite pairs and chooses which candidate demonstrates higher quality. Full training details and hyperparameters are provided in Table 1.

3.3 Win-Rate Ranking

From the 600 training samples, we construct pairwise comparisons for preference training. For each pair, the two rewrites must differ by at least 0.67 points in human score:

$$\mathcal{P} = \{(r_i, r_j) \mid |q_i - q_j| \geq 0.67\} \quad (6)$$

yielding 2,652 balanced comparisons (1,326 per class). Because each source has one rewrite, pairs are constructed across different source texts. The model receives both source–rewrite pairs and chooses which demonstrates higher overall quality relative to its own source.

At inference, all $\binom{129}{2} = 8,256$ evaluation pairs are scored; each rewrite’s *win rate* is:

$$\text{WinRate}(r_i) = \frac{1}{N-1} \sum_{j \neq i} \mathbf{1}[\text{Pref}(r_i, r_j) = 1] \quad (7)$$

Spearman correlation is then computed between win rates and human average scores. This aggregation provides a more robust quality estimate than any single absolute prediction, as each rewrite’s ranking is informed by multiple comparisons. Because each source currently has one rewrite, our main pairwise setting is cross-source global quality ranking rather than same-source candidate selection.

4 Experimental Setup

We evaluate on the 129 held-out Chinese source–rewrite pairs described in Section 3.1. We compare two training paradigms—absolute-score SFT and pairwise preference training—on two backbone models: Qwen3 8B (Yang et al., 2025) and Qwen2.5 7B Instruct (Team, 2025), both fine-tuned with LoRA. Baselines include zero-shot LLMs (Qwen2.5 72B, Qwen3 14B, Qwen2.5 14B, DeepSeek V3, GPT-4.1, and others), the general-purpose evaluator Prometheus 2 (Kim et al., 2024), G-Eval (Liu et al., 2024), and seven source-similarity metrics (BLEU, ROUGE-L, SBERT cosine, TF-IDF cosine, Jaccard word, Jaccard character, W2V cosine).

We use Spearman correlation with human average scores as the primary metric because our goal is ranking alignment rather than calibrated score prediction. Full results with Pearson r , Kendall τ , MAE, RMSE, and agreement rates are provided in Appendix A.

Table 1 summarizes the complete training configuration.

5 Results and Analysis

5.1 Main Results: Pairwise Preference vs. Absolute SFT

Table 2 presents the central comparison. Cross-source pairwise preference training achieves $\rho = 0.688$ on Qwen3 8B, outperforming absolute-score SFT ($\rho = 0.645$) by 0.043 in Spearman ρ . The same pattern holds on Qwen2.5 7B ($\rho = 0.665$ vs. $\rho = 0.580$, a

Table 1: Complete training configuration for RewriteJudge.

Hyperparameter	Value
Base model	Qwen3 8B, Qwen2.5 7B Instruct
Quantization	bf16 (no quantization)
LoRA rank (r)	16
LoRA scaling (α)	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Learning rate	2×10^{-4}
LR scheduler	Cosine (warmup 3%)
Epochs	3
Batch size	4 (gradient accumulation: 4)
Effective batch size	16
Precision	bf16
Gradient checkpointing	Enabled
Training samples	1,008 (168 per class)
GPU	1× NVIDIA H20 GPU (96GB)
Training time	~16 minutes

Table 2: Main comparison: pairwise preference training outperforms absolute-score SFT and zero-shot judges on Chinese rewriting evaluation.

Method	Model	Supervision	Spearman ρ
Pairwise (ours)	Qwen3 8B	Cross-source pref.	+0.688
Pairwise (ours)	Qwen2.5 7B	Cross-source pref.	+0.665
Absolute SFT (ours)	Qwen3 8B	Score regression	+0.645
Absolute SFT (ours)	Qwen2.5 7B	Score regression	+0.580
Zero-shot	Qwen2.5 72B	None	+0.406
Zero-shot	DeepSeek V3	None	+0.391
Gen. purpose	Prometheus 2	Multi-task	+0.145
Zero-shot	Qwen3 8B	None	+0.106

gain of 0.085). Both trained variants substantially exceed zero-shot and general-purpose judges, where even 685B-parameter models fail to reach positive correlation in some cases. Since the absolute gain on Qwen3 8B is modest, we report it as a consistent numerical improvement and further analyze why pairwise training provides a more suitable learning signal. The full comparison across all 19 methods is provided in Appendix A.

5.2 Why Pairwise Preference Helps

Pairwise supervision is data-efficient. Table 3 shows pairwise performance with different training data fractions. Qwen3 8B reaches $\rho = 0.611$ with only 25% of the data (663 pairs) and $\rho = 0.669$ with 50%, demonstrating that pairwise supervision is data-efficient and maintains stable performance even with limited training data. The Qwen2.5 7B result at 50% reveals a failure mode specific to pairwise evaluation: the model develops a strong preference for the second presented candidate,

Table 3: Pairwise data efficiency. Qwen3 8B improves steadily with more comparisons, while Qwen2.5 7B shows a position-bias failure at 50% (LoRA $r = 16$, 3 epochs).

Model	Training Data	Pairs	Spearman ρ
Qwen2.5 7B	25%	663	+0.434
	50%	1,326	+0.140
	100%	2,652	+0.665
Qwen3 8B	25%	663	+0.611
	50%	1,326	+0.669
	100%	2,652	+0.688

Table 4: Absolute-score SFT learning curve. Absolute scoring improves with more balanced examples but remains below pairwise preference training at full data.

Balanced Samples	Qwen2.5 7B	Qwen3 8B
50	−0.026	+0.058
100	+0.094	+0.117
200	+0.259	+0.136
400	+0.235	+0.247
1,008 (full)	+0.580	+0.645

which collapses the win-rate distribution after order averaging. Position swap averaging mitigates this bias by presenting each pair in both orderings; a model that always predicts the second rewrite as better produces average preference ≈ 0.5 per pair, collapsing the win rate distribution and yielding near-zero Spearman correlation. This result shows that pairwise training is not merely a way to create more training examples; it also requires explicit control of position bias. With the full comparison set, the same model recovers to $\rho = 0.665$, suggesting that sufficient preference coverage and order balancing are both necessary for stable ranking.

Absolute scoring requires more calibration data. Table 4 and Figure 3 show the absolute-score SFT learning curve. Performance improves from 50 to 1,008 samples, with a clear jump after 400 samples. With only 100 balanced samples, RewriteJudge reaches $\rho = 0.117$, already above all zero-shot baselines and general-purpose evaluators. Even at the full 1,008 balanced samples, absolute scoring remains below pairwise preference training ($\rho = 0.645$ vs. $\rho = 0.688$). This supports our central motivation: direct score prediction is a global calibration problem, whereas pairwise training can learn useful relative preferences earlier.

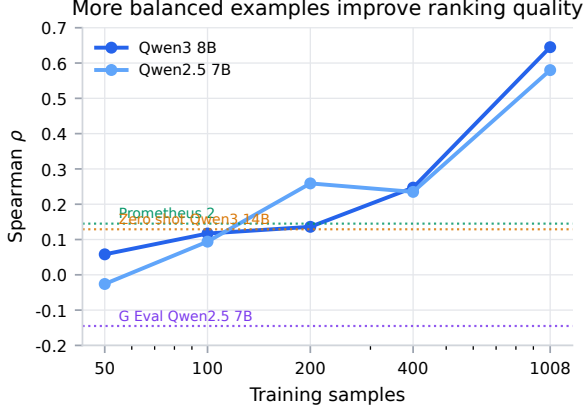


Figure 3: Full learning curve including both LoRA variants and baseline methods (dashed lines). All baselines show flat or negative correlation regardless of data size.

Table 5: LoRA rank ablation: effect on trainable parameters and Spearman correlation. We use $r = 16$ as the pre-specified setting; $r = 32$ is reported only as sensitivity analysis.

Model	Rank r	α	Params	Spearman ρ
Qwen2.5 7B	8	16	$\sim 0.1\%$	+0.685
	16	32	$\sim 0.2\%$	+0.665
	32	64	$\sim 0.4\%$	+0.705
Qwen3 8B	8	16	$\sim 0.1\%$	+0.679
	16	32	$\sim 0.2\%$	+0.688
	32	64	$\sim 0.4\%$	+0.713

Training robustness. Pairwise preference learning is also not highly sensitive to adapter capacity. Table 5 shows that changing the LoRA rank from 8 to 32 changes Spearman correlation only within a narrow range ($\Delta\rho \leq 0.03$). Since $r = 32$ is evaluated only as a sensitivity check and not used for model selection, we keep $r = 16$ as the main setting.

Ranking consistency beyond pairwise accuracy. A natural expectation is that higher pairwise accuracy should produce higher ranking correlation. Our results contradict this (Table 6): Qwen3 235B achieves 72.0% pairwise accuracy yet only $\rho = 0.132$, while our fine-tuned 8B model reaches $\rho = 0.688$ at 69.5% accuracy. This suggests that zero-shot models may correctly identify easy pairs with large quality gaps but fail to maintain consistent logic across the full spectrum. Our pairwise training produces a model that forms a stable global ranking, not merely a pair classifier.

Calibration collapse in absolute scor-

Table 6: Pairwise accuracy and ranking correlation are poorly aligned for zero-shot models.

Model	Accuracy (%)	Spearman ρ	Paradox?
Ours (8B LoRA)	69.5	+0.688	No
Ours (7B LoRA)	90.0	+0.665	No
Qwen3 235B	72.0	+0.132	Yes
Qwen2.5 72B	69.3	+0.406	Mild
DeepSeek V3	67.7	+0.391	No
Kimi K2	31.6	-0.472	No

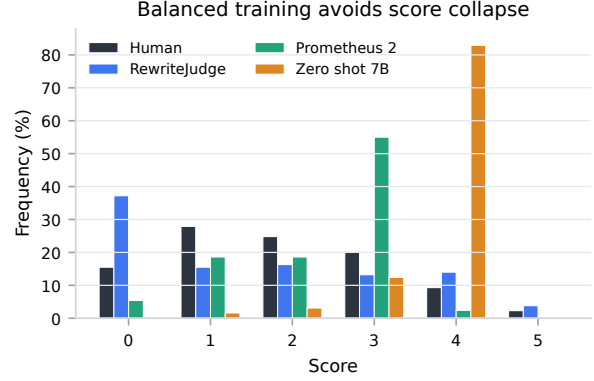


Figure 4: Distribution of human annotations and evaluator predictions. Zero-shot models concentrate predictions at Score 4; Prometheus 2 collapses to Score 3; RewriteJudge produces a distribution closer to human annotations.

ing. Figure 4 shows the score frequency across different evaluators. Zero-shot 7B has a ceiling effect: more than 80% of its predictions are Score 4. Prometheus 2 mostly predicts Score 3 and ignores much of the lower and upper score range. In contrast, RewriteJudge, for both Qwen3 8B and Qwen2.5 7B, produces a distribution closer to the human scores across all six score levels. Zero-shot and general-purpose judges often concentrate predictions in a narrow score range, whereas task-specific training produces a score distribution closer to human annotations.

To make this collapse more explicit, Figure 5 shows that Prometheus 2 predicts Score 3 for the majority of samples across nearly all human score levels. This behavior is not a small calibration offset; it is a loss of discriminative power that explains its low Spearman correlation ($\rho = 0.145$).

In contrast, both RewriteJudge variants exhibit a strong diagonal trend consistent with human judgment. The Qwen3 8B variant (Figure 6) correctly identifies 60% of Score 0 samples and 66.7% of Score 5 samples, with very

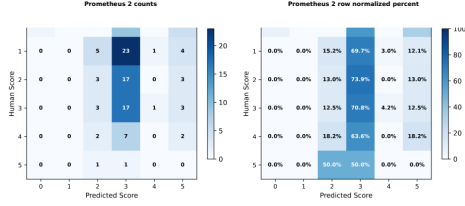


Figure 5: Confusion matrix for Prometheus 2. The model predicts Score 3 for most samples across all human score categories, exhibiting discriminative collapse.

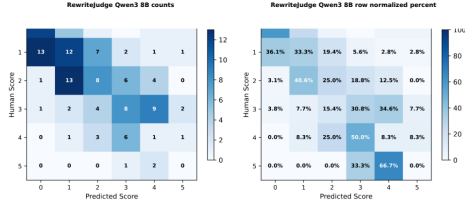


Figure 6: Confusion matrix for RewriteJudge (Qwen3 8B). The model shows a strong diagonal pattern with very few extreme misclassifications.

few extreme misclassifications. The Qwen2.5 7B variant (Figure 7) similarly concentrates predictions along the diagonal, with 50% accuracy for Score 0 and 58.3% for Score 4. Both variants rarely commit extreme errors, confirming that class-balanced fine-tuning preserves ordinal structure.

5.3 Robustness and Diagnostic Checks

Bias analysis. A reliable evaluator should judge quality based on content, not spurious features such as text length or verbosity. Table 7 compares bias across methods. Both RewriteJudge variants show minimal output length bias (Qwen3 8B: $r = -0.033$; Qwen2.5 7B: $r = -0.058$), well below zero-shot Qwen3 8B ($r = 0.325$), character overlap ($r = 0.306$), and a length heuristic ($r = 0.287$). RewriteJudge Qwen3 8B achieves near-zero verbosity bias ($\rho = 0.006$), far below zero-shot Qwen3 8B ($\rho = 0.382$) and character overlap ($\rho = 0.760$). Position bias measures candidate-order preference in pairwise evaluation; RewriteJudge maintains low position bias (Qwen3 8B: $r = 0.120$; Qwen2.5 7B: $r = 0.187$), while zero-shot Qwen3 8B shows higher bias ($r = 0.437$). Figures 8–10 provide additional visualizations of bias across dimensions.

Length bias. Figure 8 shows evalu-

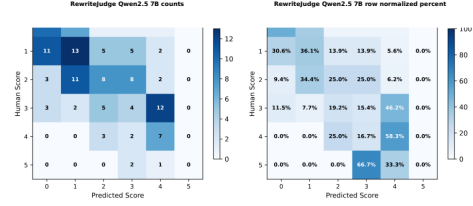


Figure 7: Confusion matrix for RewriteJudge (Qwen2.5 7B). Predictions concentrate along the diagonal, with particular strength in the middle and high score range.

Table 7: Bias analysis. Lower absolute correlation values indicate less bias. RewriteJudge shows low bias across all dimensions compared to baselines.

Method	Output Len. r	Input Len. r	Verbosity ρ	Position Bias r	Bias Score
RewriteJudge (Qwen3 8B)	-0.033	-0.070	0.006	0.120	0.057
RewriteJudge (Qwen2.5 7B)	-0.058	-0.115	0.106	0.187	0.116
Prometheus 2	0.010	-0.071	0.098	0.223	0.100
Zero-shot Qwen3 8B	0.325	0.167	0.382	0.437	0.328
Char Overlap	0.306	0.044	0.760	—	0.468
Length Heuristic	0.287	0.052	0.741	—	0.441

ator scores across output length quartiles: both RewriteJudge variants remain nearly flat, while zero-shot Qwen3 8B, character overlap, and the length heuristic all reward longer outputs regardless of quality. The length heuristic exhibits extreme verbosity bias ($\rho = 0.741$), confirming a known failure mode where naive metrics struggle to distinguish between informative expansion and redundant verbosity (Kuang et al., 2024).

Verbosity bias. Figure 9 breaks down verbosity bias by the ratio of output length to input length across two categories: Concise (ratio < 0.8) and Similar (0.8 to 1.2). RewriteJudge variants assign consistently low scores in both categories, remaining well below all other methods. Character overlap shows the largest jump ($3.4 \rightarrow 4.6$, $\Delta=1.2$), reflecting strong systematic preference for longer outputs.

Position bias. Figure 10 summarizes all four bias dimensions as a compact heatmap: both RewriteJudge variants have low overall bias scores (0.057 and 0.116), while zero-shot Qwen3 8B (0.328), character overlap (0.468), and the length heuristic (0.441) show larger bias values across multiple dimensions.

Cross-source and same-source settings. Table 8 compares the two pairwise evaluation settings. Cross-source comparisons pair rewrites from different source texts and cover a wider quality range, while same-source preference directly addresses the practical task

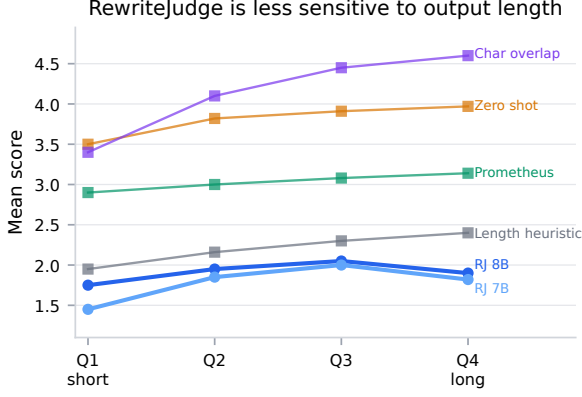


Figure 8: Mean evaluator score by output length quartile. RewriteJudge maintains consistent scores across quartiles, while traditional metrics show increasing scores for longer outputs.

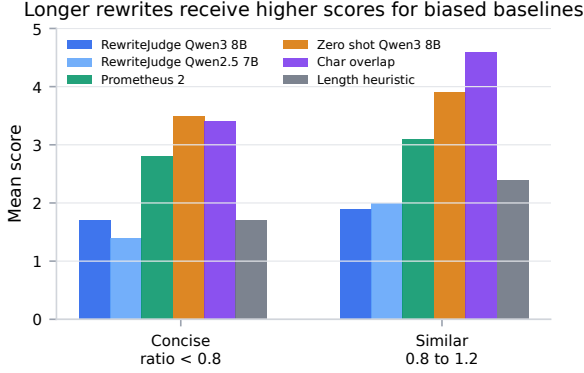


Figure 9: Verbosity bias comparison. RewriteJudge shows almost no verbosity bias, while the length heuristic exhibits extreme preference for verbose outputs ($\rho=0.741$).

of choosing among rewrites for the same input. The gap clarifies the scope of our contribution: the current method is most effective for learning a global notion of rewrite quality, while same-source selection requires denser candidate-level preference data and remains an important future direction.

Source similarity is not a reliable proxy. Before training task-specific judges, we ask whether automatic metrics based on source similarity can serve as proxies for rewrite quality. Figure 11 shows that all seven source-similarity metrics are negatively correlated with human judgment ($\rho = -0.28$ for SBERT COSINE to $\rho = -0.60$ for JACCARD CHAR), with bootstrap 95% confidence intervals that do not cross zero. Table 9 illustrates the two systematic error patterns behind this failure: close copies receive high overlap scores

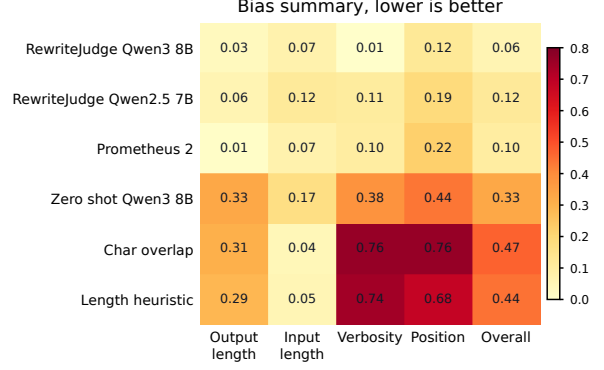


Figure 10: Bias heatmap across evaluator methods. Lower values indicate weaker correlation with spurious factors such as length, verbosity, or position.

Table 8: Cross-source and same-source preference settings on Qwen2.5 7B. Cross-source comparison is stronger for global quality ranking; same-source candidate selection remains more challenging.

Setting	Spearman ρ	Scope
Cross-source	+0.665	Global quality ranking across different inputs
Same-source	+0.421	Practical candidate selection; harder and less solved

but low human ratings, while strong rewrites that substantially transform the text receive low overlap scores despite high human quality. Additional error analysis with concrete examples is provided in Appendix B.

6 Conclusion

We studied how to train LLM-based judges for Chinese text rewriting evaluation, comparing absolute-score SFT with pairwise preference training. Using 730 human-rated Chinese source-rewrite pairs, we showed that pairwise preference training achieves the highest ranking alignment ($\rho = 0.688$), outperforming absolute-score SFT ($\rho = 0.645$) and substantially exceeding zero-shot and general-purpose LLM judges. Analyses show that pairwise supervision is data-efficient, produces better ranking consistency beyond raw pairwise accuracy, and exhibits low bias across length, verbosity, and position dimensions. Future work should investigate same-source candidate selection, larger human validation, and calibrated score prediction.

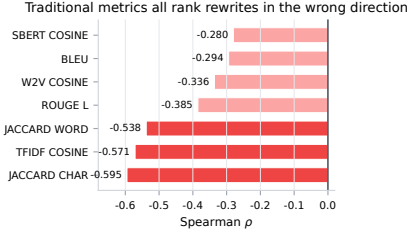


Figure 11: Source-similarity metrics are negatively aligned with human rewrite quality. All seven metrics show negative Spearman ρ with bootstrap 95% CIs that do not cross zero.

Table 9: Two representative failure modes of source-similarity metrics.

Case	BLEU	Human	Error mode
Minor lexical substitution	0.55	0.7	High overlap rewards copying
Substantial faithful rewriting	0.00	4.0	Low overlap penalizes useful transformation

7 Limitations

Our dataset contains 730 Chinese general-purpose improvement pairs from a single-rewrite-per-source design; generalization to other languages, rewriting subtasks (style transfer, simplification), and same-source candidate selection remains unknown.

Cross-source pairwise aggregation achieves the highest correlation ($\rho = 0.688$), but applies to global ranking rather than same-source selection and comes at $O(n^2)$ cost.

Our analysis focuses on ranking quality; better-calibrated numeric scores are an open direction. Full-parameter tuning and larger annotated datasets may further improve performance beyond the LoRA adaptation used here. This result demonstrates that task-specific supervision helps for this benchmark and task definition, not that RewriteJudge generalizes as a universal rewriting evaluator.

References

Tanvir Alam, Minwoo Ryu, Immanuel Adame, and Gholamreza Haffari. 2024. A generalized framework for paraphrase quality evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14070–14084.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Siyuan Zhuang, Zhilin Zhang, Yi Zhuang, Joseph E Gonzalez, Ion Stoica, and Eric P Xing. 2024. Chatbot arena: An open platform for

evaluating LLMs by human preference. *arXiv preprint arXiv:2403.04132*.

Mingqi Gao, Xinyu Hu, Jie Ruan, Xiao Pu, and Xiaojun Wan. 2025. LLM-based NLG evaluation: Current status and challenges. *Computational Linguistics*, 51(2):661–687.

Sebastian Gehrmann, Elizabeth Clark, and Thibault Sellam. 2023. Repairing the cracked foundation: A survey of obstacles in evaluation practices for generated text. *Journal of Artificial Intelligence Research*, 77:103–166.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Seungone Kim, Joongbo Shin, Joel Jang, Ha-neul Xu, Sungdong Suh, and 1 others. 2024. Prometheus 2: An open source language model specialized in evaluating other language models. *arXiv preprint arXiv:2405.01535*.

Jiaxin Kuang, Yue Xu, Yu Zhu, and Bill Yuchen Lin. 2024. Evaluating long-form text generation: A theme of biases. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 8878–8898.

Nathan Lambert, Valentina Pyatkin, Jacob Morrison, Lester James Miranda, Bill Yuchen Lin, Khyathi Chandu, and 1 others. 2024. Reward-Bench: Evaluating reward models for language modeling. In *arXiv preprint arXiv:2403.13787*.

Wuwei Lan and Yiming Chen. 2023. ParaScore: Evaluating paraphrasing with embeddings and pairwise generation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11030–11049.

Yohan Lee and 1 others. 2025. A reliable LLM-as-a-judge framework for evaluating text generation using unsupervised fine-grained annotations. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*.

Chin-Yew Lin. 2004. ROUGE: A package for automatic evaluation of summaries. *Text Summarization Branches Out*, pages 74–81.

Yang Liu, Kyoungmin Cho, James Evans, Seokhwan Yoo, Sewon Min, Luke Zettlemoyer, Hannaneh Hajishirzi, and Luke S Park. 2024. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6705–6723.

- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992.
- Qwen Team. 2025. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.
- Haotian Wang, Shuai Bai, Zhenwei Zhu, Zhiqi Wang, Jun Liu, Kaisheng Li, Ya-Qin Ren, Jieyang Chen, and 1 others. 2024a. Evaluating vision-language models on generation and grounding. In *arXiv preprint arXiv:2407.11579*.
- Peiyi Wang, Lei Li, Liang Chen, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. 2024b. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- Yuntong Wang and 1 others. 2024c. Generative reward models. *arXiv preprint arXiv:2404.03289*.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.
- Weizhe Yuan, Graham Neubig, and Pengfei Liu. 2021. BARTScore: Evaluating generated text as text generation. In *Advances in Neural Information Processing Systems*, volume 34, pages 27263–27277.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations*.
- Wei Zhao, Maxime Chen, Pranav Ananth, Luning Liang, Wen He, and Wei Peng. 2019. Mover-score: Text generation evaluating with contextualized embeddings and earth mover distance. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 563–578.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yi Zhuang, Zi Lin, Zhongyi Li, Dacheng Li, Yong Zhang, Joseph E Gonzalez, Ion Stoica, and Eric P Xing. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36.

Table 10: Full evaluation results on the Chinese rewriting quality benchmark. Source-similarity metrics are negatively correlated with human scores. Best results are in **bold**.

Method	Type	Spearman ρ	Pearson r	Kendall τ	MAE	RMSE	Exact / ± 1 / ± 2 (%)
RewriteJudge (Ours, Qwen3 8B)	Fine tuned (8B)	+0.645	+0.602	+0.530	0.91	1.24	25.6 / 79.8 / 95.4
+Reasoning prefix	Fine tuned (8B)	+0.360	+0.374	+0.312	1.07	1.36	14.0 / 72.1 / 93.0
Unbalanced training (Qwen3 8B)	Fine tuned (8B)	+0.337	+0.344	+0.289	1.12	1.43	14.0 / 70.5 / 90.7
RewriteJudge (Ours, Qwen2.5 7B)	Fine tuned (7B)	+0.580	+0.580	+0.470	0.94	1.25	24.8 / 75.2 / 93.0
Unbalanced training (Qwen2.5 7B)	Fine tuned (7B)	+0.287	+0.246	+0.238	1.10	1.40	14.0 / 72.1 / 91.5
Prometheus 2	Fine tuned (7B)	+0.145	+0.126	+0.120	1.54	1.95	17.0 / 53.8 / 75.5
Zero shot (Qwen3 14B)	Zero shot (14B)	+0.129	+0.180	+0.112	2.03	2.37	10.9 / 35.7 / 56.6
Zero shot (Qwen2.5 14B)	Zero shot (14B)	+0.058	+0.105	+0.046	2.14	2.49	11.6 / 30.2 / 56.6
Zero shot (Qwen3 8B)	Zero shot (8B)	+0.106	+0.176	+0.093	1.86	2.23	15.5 / 40.3 / 62.8
Zero shot (Qwen2.5 7B)	Zero shot (7B)	-0.019	+0.079	-0.016	2.04	2.38	10.1 / 37.2 / 56.6
G-Eval (Qwen3 8B)	Zero shot (8B)	+0.041	+0.089	+0.033	1.85	2.22	16.3 / 41.1 / 58.1
G-Eval (Qwen2.5 7B)	Zero shot (7B)	-0.145	-0.064	-0.121	2.22	2.62	12.4 / 33.3 / 52.7
SBERT COSINE	Traditional	-0.280	-0.072	-0.199	NA	NA	NA
BLEU	Traditional	-0.294	-0.211	-0.243	NA	NA	NA
W2V COSINE	Traditional	-0.336	-0.099	-0.238	NA	NA	NA
ROUGE L	Traditional	-0.385	-0.396	-0.317	NA	NA	NA
JACCARD WORD	Traditional	-0.538	-0.509	-0.416	NA	NA	NA
TFIDF COSINE	Traditional	-0.571	-0.503	-0.433	NA	NA	NA
JACCARD CHAR	Traditional	-0.595	-0.550	-0.453	NA	NA	NA

A Full Baseline Results

Table 10 provides the full comparison across 19 methods. Source-similarity metrics are all negatively correlated with human scores ($\rho = -0.28$ to $\rho = -0.60$). Among zero-shot LLMs, Qwen2.5 72B ($\rho = 0.406$) and DeepSeek V3 ($\rho = 0.391$) show moderate positive correlation, while several larger or newer models perform poorly: GPT-4.1 ($\rho = -0.145$), DeepSeek V3.2 ($\rho = -0.274$), Kimi K2 ($\rho = -0.472$). Prometheus 2, despite being trained for evaluation, reaches only $\rho = 0.145$, confirming that general-purpose evaluator training does not transfer to Chinese rewriting.

Accuracy vs. ranking correlation. As visualized in Figure 12, Qwen3 235B achieves high zero-shot accuracy (72.0%) yet fails to provide a meaningful ranking ($\rho = 0.132$). This suggests that zero-shot models may correctly identify easy pairs with large quality gaps but fail to maintain consistent logic across the full spectrum. Our fine-tuned model variants largely avoid this mismatch, as seen by their position in the upper right area of the figure.

Zero-shot model scale. Figure 13 shows within-family zero-shot comparisons. These results show that newer or larger endpoints are not reliably better in our prompting setup, but should not be read as a general generational trend.

Format compliance vs. evaluation quality. Table 11 shows the parse failure rate for each zero-shot baseline. The results in Fig-

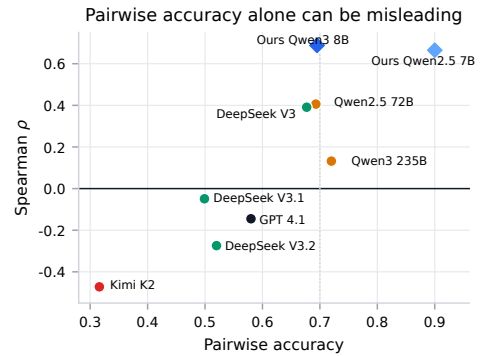


Figure 12: Pairwise accuracy vs. Spearman ρ : high accuracy but low ρ shows that accuracy can mislead for zero-shot evaluators.

ure 14 confirm that format compliance and evaluation quality are **orthogonal**. Models with 100% compliance (0% failure) span the entire performance spectrum, from the best zero-shot baseline (DeepSeek V3) to the worst (Kimi K2). This indicates that the mechanical ability to follow instructions is independent of the critical capacity to judge text quality.

Table 11: Format compliance vs. evaluation quality. Models with zero parse failures include both the best and worst performers, indicating format compliance is independent of evaluation capability.

Model	Parse Fail (%)	Spearman ρ	Quality
DeepSeek V3	0.0	+0.391	Best zero-shot
Kimi K2	0.0	-0.472	Worst zero-shot
DeepSeek V3.1	0.0	-0.049	Poor
DeepSeek V3.2	0.0	-0.274	Poor
GPT 4.1	0.0	-0.145	Poor
Qwen2.5 72B	8.8	+0.406	Good
Qwen3 235B	9.0	+0.132	Poor

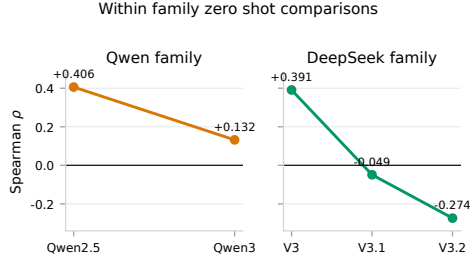


Figure 13: Within-family zero-shot comparisons. These results show that newer or larger endpoints are not reliably better in our prompting setup, but should not be read as a general generational trend.

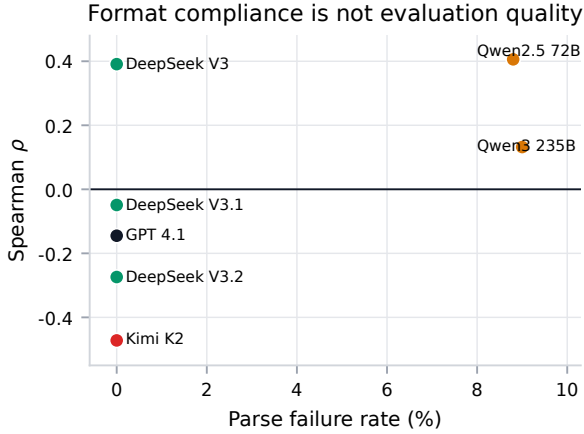


Figure 14: Parse failure rate vs. Spearman ρ . No correlation exists between format compliance and evaluation quality: models with 0% parse failures span the full quality range.

B Source-Similarity Error Analysis

Section 5 summarizes the core error mechanism in the main text. Here we provide additional detail on the two recurring patterns behind the negative correlation between overlap metrics and human judgment.

Pattern 1: Surface similarity without quality. Rewrites that are close copies of the source text, changing only a few words while preserving the original structure, receive high overlap metric scores but low human ratings. For example, a rewrite that substitutes “加盟” with “加盟后” and “场场” with “几乎每场” achieves BLEU=0.55 but receives a human score of only 0.7. The metric rewards lexical overlap that reflects copying, not genuine rewrite quality.

Pattern 2: Quality without surface similarity. Strong rewrites that restructure the sentence, using different vocabulary and syntax while preserving meaning, receive very

low metric scores but high human ratings. A rewrite that restructures a passage about a press conference into a more concise summary achieves BLEU=0.00 but receives a human score of 4.0. The metric penalizes the very lexical variation that characterizes successful rewriting.

These two patterns show that overlap metrics and human ratings answer different questions: overlap metrics measure how *similar* a rewrite is to its source, while human judgment evaluates how *good* the rewrite is. The critical insight is that these are not merely different metrics; they are *inversely correlated* for rewriting. Pattern 1 shows that high source similarity indicates *insufficient transformation* (the rewrite barely changed anything), while Pattern 2 shows that high source dissimilarity indicates *successful transformation* (the rewrite meaningfully transformed the text). This inverse relationship explains why the negative correlations are universal and statistically significant rather than reflecting random noise or a specific metric deficiency. For text rewriting, a task where the goal is to differ from the source while preserving meaning, these two objectives are inversely correlated. This explains why the strongest negative correlations are observed for character level metrics (JACCARD CHAR: $\rho=-0.60$), which most directly measure surface overlap, while semantic metrics like SBERT COSINE ($\rho=-0.28$) show somewhat weaker, but still negative, correlations.

Chinese annotation examples. Table 12 presents several representative cases covering various quality levels. These examples illustrate the nuances of Chinese text rewriting, ranging from simple lexical substitutions to sophisticated stylistic enhancements. We observe that REWRITEJUDGE (RJ) predictions align closely with human gold standard scores, successfully capturing subtle improvements in fluency and expressiveness while remaining sensitive to instances where the rewrite fails to provide significant added value.

C Annotation Guidelines and Prompts

We investigate how prompt detail affects evaluation quality by comparing three variants:

Table 12: Representative examples from the evaluation benchmark. **Source**: original text; **Rewrite**: model generated rewrite; **Human**: average human score; **RJ**: RewriteJudge prediction.

Source	Rewrite	Human	RJ
今天天气很好，我们决定去公园散步。	阳光明媚，微风轻拂，我们来到公园漫步，享受这惬意的时光。	4	4
这个问题的解决方案需要进一步研究。	要解决此事还需深入探究才能找到答案。	2	2
他是一个非常好的学生，学习成绩优异。	他学习刻苦努力，是个品学兼优的好学生。	3	3
这家餐厅的菜很好吃。	这儿的菜品味道鲜美可口，令人回味无穷。	3	2
会议上讨论了关于环保的问题。	大家在会议中就环境保护议题展开了深入探讨。	4	4

Simple (minimal), **Standard** (brief criteria), and **Detailed** (detailed rubrics). The **Detailed** prompt exactly matches our benchmark’s human rubric, grounding the model directly in the human annotation protocol.

Simple prompt

Simple prompt	
请帮我评判输入文本和输出文本重写的质量，并给出最终综合评分（0 到 5 分的整数）。	<i>Please evaluate the rewriting quality between the input and output texts, and provide a final overall score (integer from 0 to 5).</i>

Standard prompt

Standard prompt	
你是一个专业的中文文本改写质量评估专家。请根据以下维度对中文文本改写质量进行评分（0 到 5 分）： 评分维度： - 语义一致性：改写是否保留了原文的核心语义 - 句式重构：改写是否进行了句法结构的改变 - 词汇变化：改写是否使用了不同的词汇表达 - 风格保持：改写是否保持了原文的风格和长度 评分标准： 0 分：改写完全失败（严重语义扭曲或毫无改写） 1 分：改写质量很差 2 分：改写质量较差 3 分：改写质量一般 4 分：改写质量较好 5 分：改写质量优秀 请先简要分析，然后给出最终综合评分（0 到 5 分的整数）。	<i>You are a professional expert in evaluating Chinese text rewriting quality. Please score the rewriting quality from 0 to 5 along the following dimensions:</i> <i>Evaluation dimensions:</i> <i>Semantic consistency: Does the rewrite preserve the core meaning of the original?</i> <i>Syntactic reconstruction: Does the rewrite alter the syntactic structure?</i> <i>Lexical variation: Does the rewrite employ different vocabulary?</i> <i>Style preservation: Does the rewrite maintain the original style and length?</i> <i>Scoring criteria:</i> <i>0: Rewriting completely failed (severe semantic distortion or no change at all)</i> <i>1: Very poor rewriting quality</i> <i>2: Poor rewriting quality</i> <i>3: Moderate rewriting quality</i> <i>4: Good rewriting quality</i> <i>5: Excellent rewriting quality</i> <i>Please briefly analyze first, then provide a final overall score (integer from 0 to 5).</i>

Detailed prompt

Detailed prompt: identical to human annotation rubric	
请帮我判断输入文本与输出文本是否满足以下要求： • 要求 1. 在保持原文含义不变的前提下，用原文不同的文字进行表达，不额外新增、删减会严重影响语义的内容。 • 要求 2. 在保证逻辑正确、语序通顺的前提下，变换实体、短语、概念出现的前后顺序、结构关系；顺序变换幅度越小，分数越低。 • 要求 3. 在保证特殊实体、引用、解释、特定语句不变的前提下，使用同义词替换；同义词替换越少，分数越低。 • 要求 4. 保持与原文语言风格一致，字数长度变化合理；语言风格差距越大，分数越低。 • 要求 5. 改写综合评分要非常严格，同时考虑上述 4 个要求；在保证语义一致（要求 1）的前提下，（要求 2,3）变化越大越好，字数越接近越好（要求 4）；如果存在大量长字段完全复制，则给 0 分。 评分准则，每个要求最低 0 分，满分均为 5 分，评分要尽可能严格。	<i>Please evaluate whether the output text satisfies the following requirements with respect to the input text:</i> • Req. 1. <i>Express the same meaning as the original using different wording, without adding or omitting content that significantly affects semantics.</i> • Req. 2. <i>Rearrange the order and structural relations of entities, phrases, and concepts while ensuring logical correctness and fluency; the smaller the reordering, the lower the score.</i> • Req. 3. <i>Replace words with synonyms while keeping special entities, citations, explanations, and fixed expressions unchanged; fewer synonym substitutions yield a lower score.</i> • Req. 4. <i>Maintain a language style consistent with the original and a reasonable change in length; the greater the stylistic divergence, the lower the score.</i> • Req. 5. <i>Apply strict holistic scoring that simultaneously considers all four requirements above. Given semantic consistency (Req. 1), maximize structural and lexical changes (Reqs. 2 and 3) while keeping length close to the original (Req. 4). If large verbatim spans are copied, assign a score of 0.</i> <i>Scoring guideline: each requirement is scored from 0 (minimum) to 5 (maximum); scoring should be as strict as possible.</i>